TDWG Standard · GBIF & OBIS · Open Biodiversity Data

# Darwin Core (DwC), Data Publishing, Citation and IPT

Mobilizing, standardizing, and crediting global biodiversity records across the Open Science research lifecycle.

Begin Lecture →

Open Science Course

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# The Problem: The Biodiversity Tower of Babel

Before standardization, natural history collections and field scientists recorded data in disparate silos:

### 🏷️ Taxonomy Chaos

Synonyms, typos, regional vernacular names, and conflicting authorities without persistent identifiers.

### 🗺️ Spatial Ambiguity

"Behind the old mill" vs degrees/minutes/seconds vs local UTM grids; lack of coordinate uncertainty metrics.

### 📅 Temporal Inconsistency

`04/05/06` (May 4th or April 5th?), arbitrary date intervals, time zones missing.

🎯 **The Consequence:** Global synthesis (climate change range shifts, extinction tracking, invasive species modeling) was virtually impossible without manual data wrangling.

# What is Darwin Core (DwC)?

**Darwin Core (DwC)** is the international standard maintained by **TDWG (Biodiversity Information Standards)**.

- **A Glossary of Terms:** A stable vocabulary with precise URI definitions (e.g. `dwc:occurrenceID`, `dwc:scientificName`).
- **Pragmatic & Flexible:** Storage-independent (flat CSV, relational SQL, JSON-LD, RDF XML).
- **The Lingua Franca of Global Portals:** The foundational exchange format for **GBIF**, **OBIS**, and national biodiversity nodes.

🔗 **Essential Reference Guide:**
Consult the official Darwin Core Quick Reference Guide (dwc.tdwg.org) for every term definition, namespace, and normative examples.

**Official TDWG Identity** tdwg.org ↗

| | |
|---|---|
| Standard: | **Biodiversity Information Standards (TDWG)** |
| Ratified: | October 2009 (Living standard) |
| Guide URL: | dwc.tdwg.org |
| Namespace: | http://rs.tdwg.org/dwc/terms/ |
| Quick List: | dwc.tdwg.org/list/ |

*"Darwin Core is primarily based on taxa, their occurrence in nature as documented by observations, specimens, samples, and related information."*

# What is GBIF?

The **Global Biodiversity Information Facility** (GBIF.org) is an international network and research infrastructure funded by governments worldwide.

### The World's Largest Biodiversity Aggregator

GBIF provides free and open access to biodiversity data globally, mobilizing evidence from museum specimens, ecological surveys, eDNA, and citizen science (e.g. iNaturalist, eBird).

**Global Open Mandate:** Standardized open licenses (CC0, CC-BY, CC-BY-NC) and citable DOIs for all datasets and search queries.

**Powered by Darwin Core:** Universal lingua franca harmonizing disparate records into a single queryable global index.

**Conservation & Policy:** Essential evidence for IUCN Red List assessments, IPBES reports, and biodiversity targets.

**2.9+ Billion**
**Species Occurrence Records**
Museum specimens, citizen science, eDNA, sensors

**2,500+**
**Publishing Institutions**
Universities, museums, government agencies, NGOs

**100+**
**Voting Participants & Nodes**
Country delegations and international organizations

# How GBIF Operates: Nodes & Governance

GBIF is **not an uncurated upload bucket** — it is a **federated, endorsed network**:




### Country / Participant Nodes

Designated national teams (e.g. GBIF Norway, GBIF Spain) serving as the official bridge between local institutions and the global infrastructure.

### Institutional Endorsement

**Strict vetting:** Node managers must review and approve every publishing organization before its datasets can be registered on GBIF.

### Federated Publishing (IPT)

Organizations keep ownership of their data. The node hosts or assists with the **IPT**, which exports Darwin Core Archives for GBIF to harvest.

# GBIF Dataset Classes Overview

GBIF standardizes biodiversity publishing into **4 distinct dataset classes** (gbif.org/dataset-classes):

### 1. Occurrence

`Core: dwc:Occurrence`

Evidence of a species at a given time and place (museum specimens, citizen science observations, camera traps).

`Most common on GBIF`

### 2. Sampling-Event

`Core: dwc:Event + Occ Ext`

Standardized environmental monitoring, surveys, transects with quantifiable effort and non-detection (absence) data.

`Ecological time-series`

### 3. Checklist

`Core: dwc:Taxon`

Catalogues of taxa, national inventories, red lists, and taxonomic classifications without specific coordinate events.

`Taxonomic backbone`

### 4. Metadata-only

`Core: None (EML XML)`

Describing undigitized physical collections, historical literature, or projects awaiting electronic mobilization.

`Collection discovery`

# GBIF Recommendations: Mandatory vs. Recommended Terms

Explore terms per dataset class based on gbif.org/dataset-classes and dwc.tdwg.org:

Occurrence | Sampling-Event | Checklist | Metadata-Only

All | ● Mandatory Only

## Occurrence Datasets (Core: dwc:Occurrence)

GBIF Class Docs

*Evidence of the presence, abundance, or absence of a biological organism at a specific location and date.*

| Darwin Core Term | GBIF Status | Definition & Purpose | Example Value |
|---|---|---|---|
| **dwc:occurrenceID** Occurrence | ● Mandatory | Globally unique, persistent identifier for the occurrence record. ↳ Primary key; GBIF indexing fails without it | urn:uuid:8b3e8e2b-2e92-498c-851c-a1288c8eb25a |
| **dwc:basisOfRecord** Occurrence | ● Mandatory | Controlled vocabulary denoting the specific nature of the record. ↳ Required controlled vocabulary for filtering | HumanObservation, PreservedSpecimen, MaterialSample |

# The Anatomy of a Darwin Core Record

A practical example of how a biological observation maps directly to standardized DwC terms:

```
{
  "dwc:occurrenceID": "urn:uuid:8b3e8e2b-2e92-498c-851c-a1288c8eb25a",
  "dwc:basisOfRecord": "HumanObservation",
  "dwc:scientificName": "Bombus terrestris (Linnaeus, 1758)",
  "dwc:kingdom": "Animalia",
  "dwc:taxonRank": "species",
  "dwc:decimalLatitude": 50.0755,
  "dwc:decimalLongitude": 14.4378,
  "dwc:geodeticDatum": "WGS84",
  "dwc:coordinateUncertaintyInMeters": 15,
  "dwc:eventDate": "2024-06-12T10:15:00Z",
  "dwc:recordedBy": "Michal Torma",
  "dcterms:license": "http://creativecommons.org/publicdomain/zero/1.0/"
}
```

### Mandatory Core Terms

- `occurrenceID`
- `basisOfRecord`
- `scientificName`
- `eventDate`

### Strongly Recommended

- `decimalLatitude`
- `decimalLongitude`
- `geodeticDatum`
- `coordinateUncertaintyInMeters`

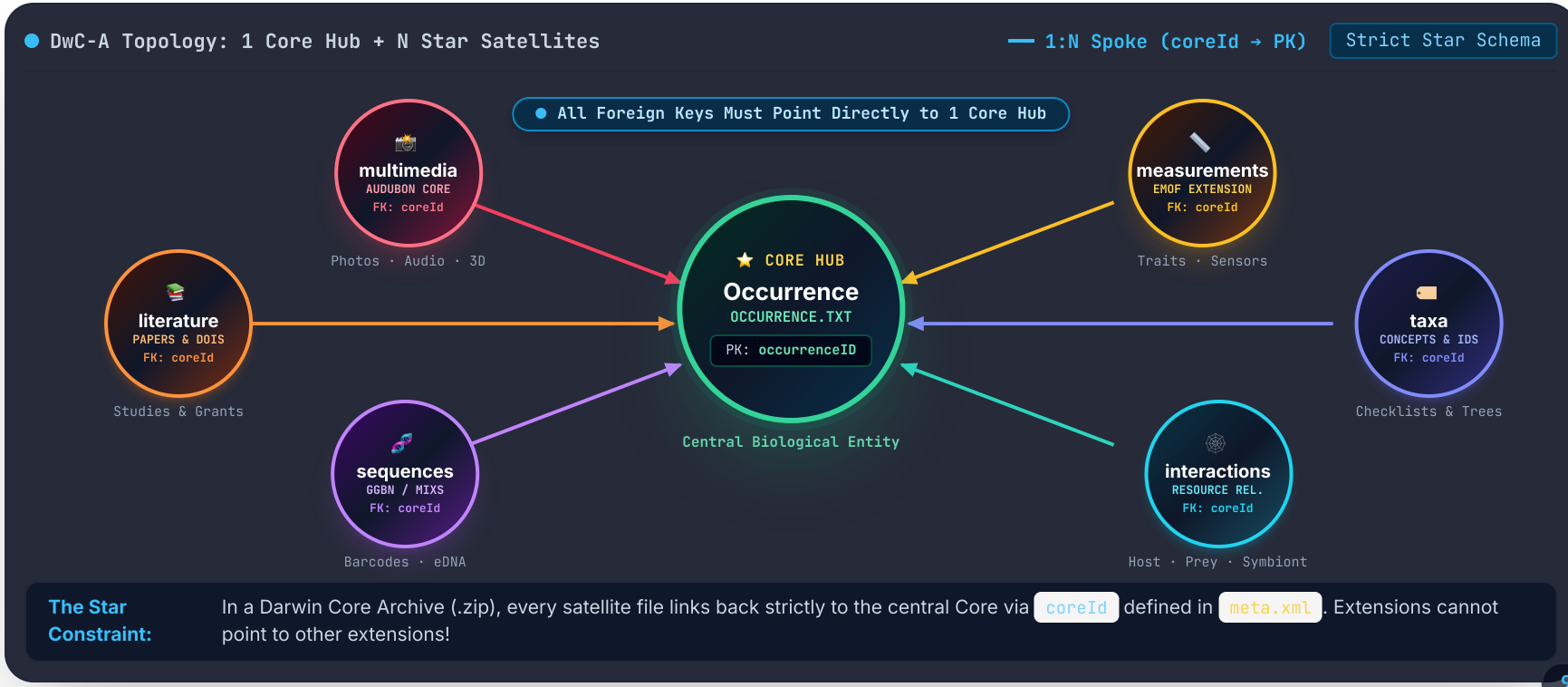
### Mandatory Open License

Machine-readable license URL is required for GBIF publication:

- `CC0 1.0`
- `CC-BY 4.0`
- `CC-BY-NC 4.0`

# The Darwin Core Star Schema (DwC-A)

A single flat table cannot cleanly model **one-to-many (1:N)** biological relationships. A **Darwin Core Archive (DwC-A)** solves this using a **Star Schema**:



# GBIF Extensions in Practice

GBIF maintains a community-curated Extension Registry to extend Core records with specialized data types:

### 📸 Simple Multimedia (Audubon Core)

Media

Attaches photos, field audio recordings, or 3D specimen scans to an occurrence record.

Terms: `identifier`, `format`, `license`

iNaturalist · Herbarium scans

### 🔪 Extended MeasurementOrFact (eMoF)

Traits & Sensors

Records quantitative traits (body length, mass) and environmental metrics (water temp, depth).

Terms: `measurementType`, `Value`, `Unit`

OBIS · Trait databases

### 🧬 DNA Derived Data (GGBN / MIxS)

eDNA & Sequences

Links DNA barcoding, metabarcoding, and environmental DNA (eDNA) detections to vouchers.

Terms: `target_gene`, `pcr_primer`, `accession`

NCBI · BOLD · ENA

### 🔗 Resource Relationship

Interactions

Documents ecological interactions between occurrences (host-parasite, pollinator, prey).

Terms: `relationshipOfResource`, `relatedResourceID`

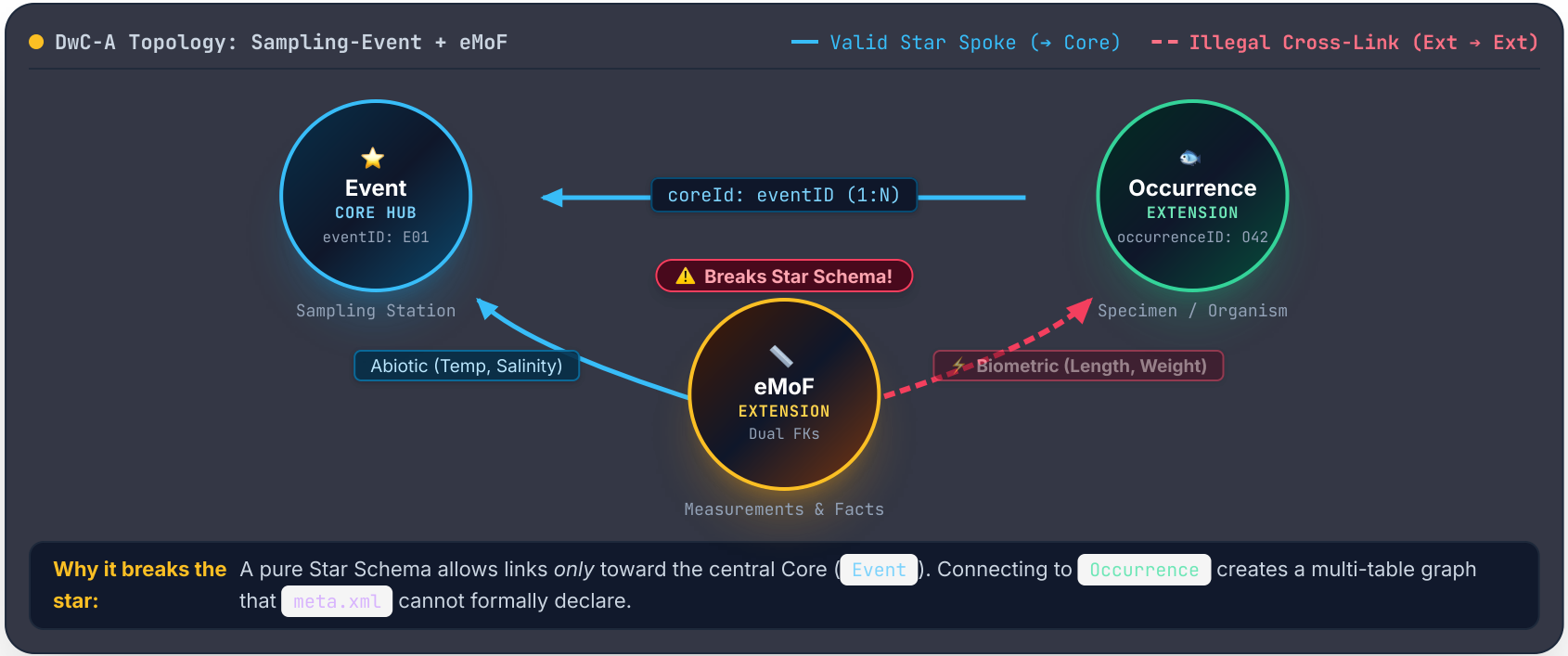
GloBI · Food webs

💡 Each extension file links back to `coreId` in `occurrence.txt` via schema defined in `meta.xml`

Explore Registry ↗

# Special Case: ExtendedMeasurementOrFact (eMoF)

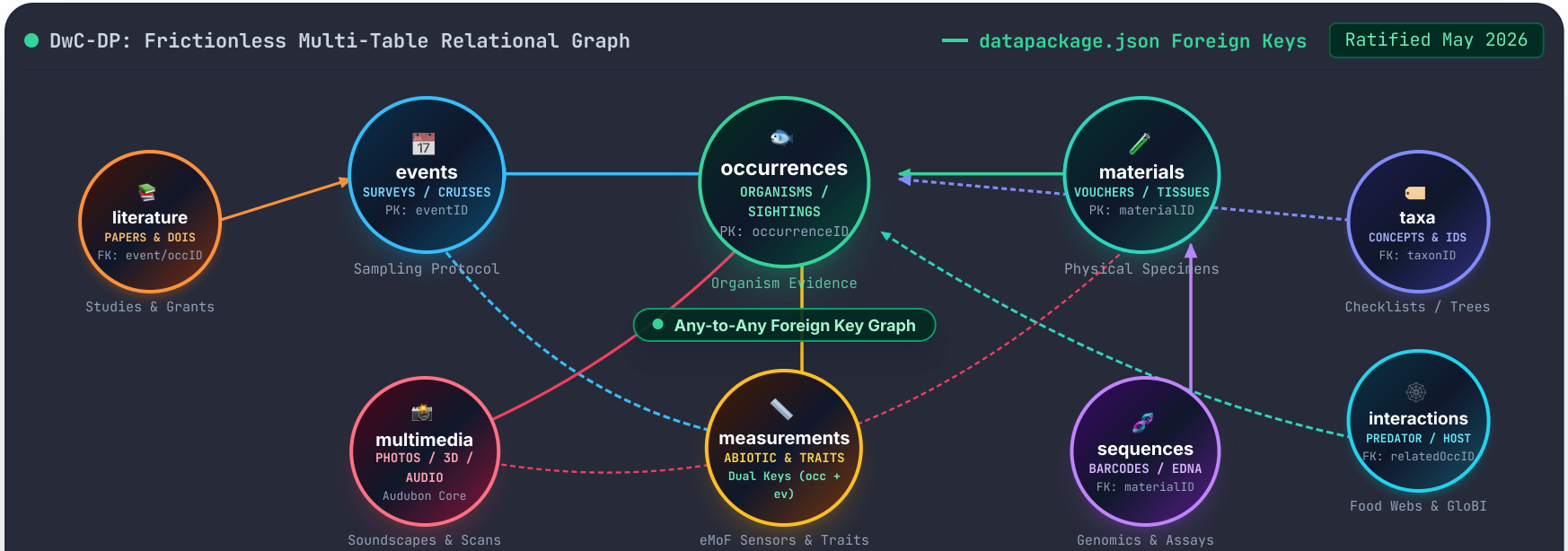
In Sampling-Event datasets, measurements target **both** the Event (abiotic) and Occurrence (biometric traits), breaking the pure Star Schema:




**Why it breaks the star:** A pure Star Schema allows links *only* toward the central Core (`Event`). Connecting to `Occurrence` creates a multi-table graph that `meta.xml` cannot formally declare.

# The Solution: Darwin Core Data Package (DwC-DP)

Officially ratified by **TDWG on May 26, 2026**: Replacing the rigid Star Schema with Frictionless Data Packages and multi-table relational graphs.





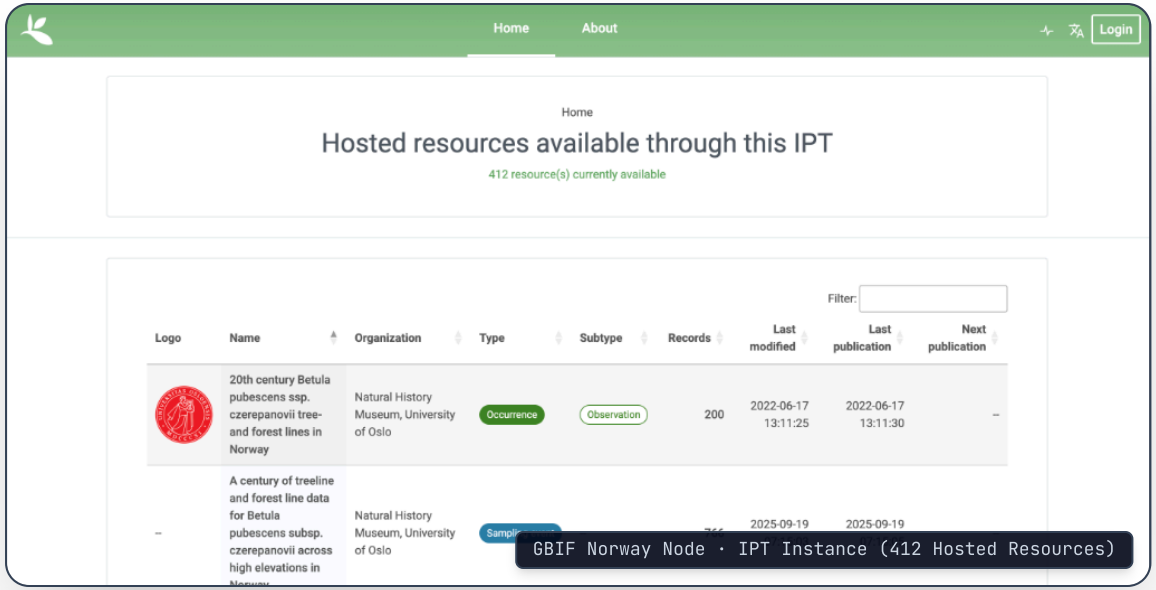
*"That looks complicated..."*

**...I hope there are tools to help me!**

Don't worry — you don't have to build these archives or schemas by hand!

# Meet the Integrated Publishing Toolkit (IPT)

GBIF's open-source web application designed to manage, document, and publish biodiversity datasets without writing raw XML or JSON.




GBIF Norway Node · IPT Instance (412 Hosted Resources)

### What is the IPT?

GBIF's free open-source publishing tool (Java web app), typically hosted by **national participant nodes** or institutions.

### Automated Packaging

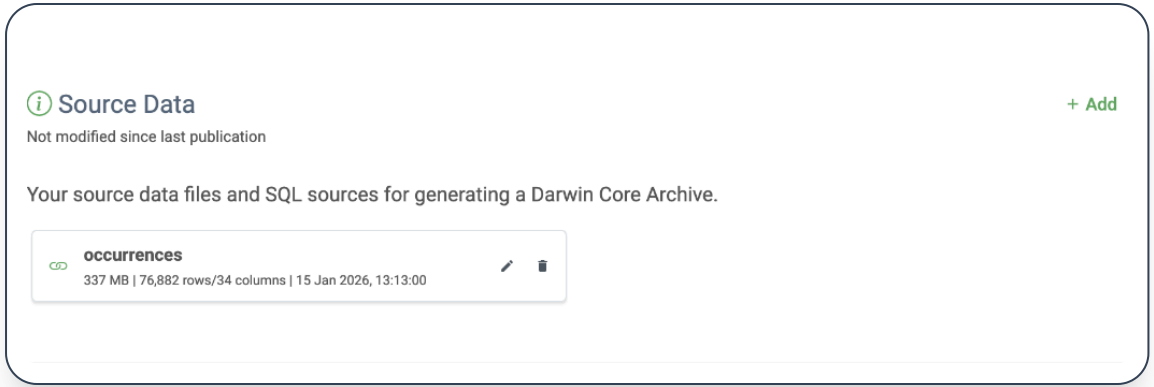
Upload CSV/Excel tables or connect databases. IPT maps your fields to **DwC terms** and packages valid archives automatically.

### Direct Node Registration

Registers datasets directly into the **GBIF Registry** with persistent DOIs and scheduled crawler sync.

# IPT Workflow · Step 1: Source Data

Connecting raw observation files or live database queries to the IPT resource.




IPT Source Data Interface: File or SQL database table connection

### 📁 Supported Formats

Delimited text files (**CSV, TSV**), Microsoft Excel spreadsheets, or direct live **SQL database connections** (PostgreSQL, MySQL, Oracle).

### ⚡ Scalability

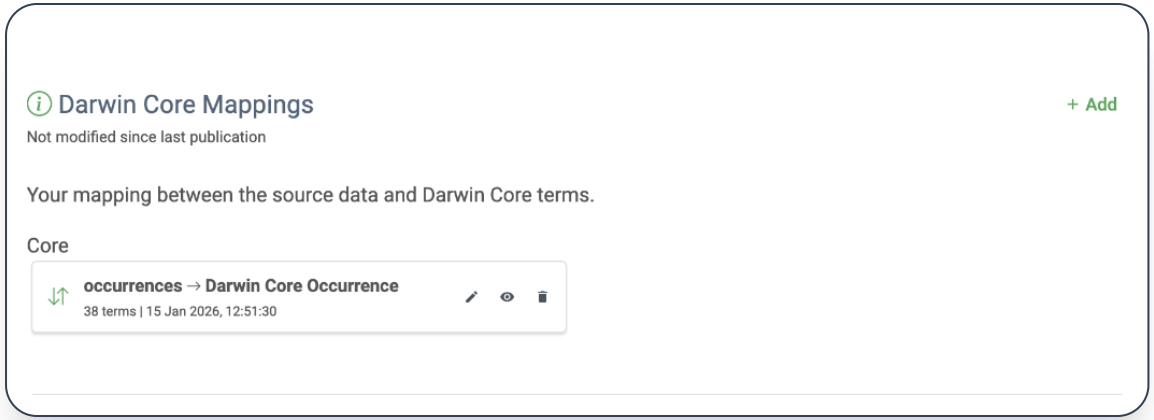
Handles small field survey lists up to millions of rows (as seen here with **337 MB / 76,882 rows** in an occurrence source).

### 🔒 Preserves Original Data

Your original column headers and internal IDs remain untouched — mapping to Darwin Core happens in the next step.

# IPT Workflow · Step 2: Darwin Core Mappings

Translating local column headers into standardized Darwin Core terms and schemas.




Core & Extension Mappings: Mapping 38 Darwin Core terms

### 🎯 Core vs. Extension Selection

Designate your main table (e.g. **Darwin Core Occurrence** or **Event**), and attach any number of registered extensions.

### 🪄 Smart Auto-Mapping

The IPT inspects column names and automatically matches standard terms (e.g. `lat` → `decimalLatitude`).

### 🔍 Value Translation & Vocabularies

Map internal codes (e.g., `OBS`) to controlled TDWG vocabularies (`HumanObservation`) directly in the UI.

# IPT Workflow · Step 3: Resource Metadata

Describing the Who, What, Where, When, and Why using Ecological Metadata Language (EML).




IPT Metadata Status: Validated EML document ready for publication

### 📋 Comprehensive EML Profile

Fills out dataset title, abstract, creators, contacts, geographic bounding boxes, taxonomic scope, and temporal ranges.

### ✅ Real-Time Validation Pill

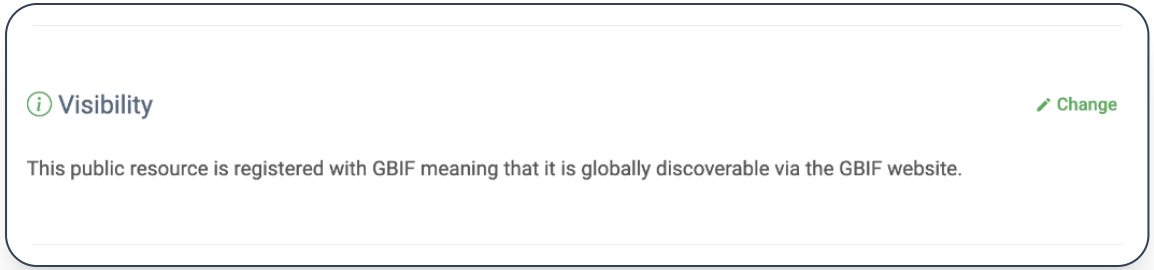
The green `Valid` badge confirms all mandatory GBIF and DataCite citation fields are complete.

### ⚖️ Machine-Readable Licensing

Select standard open licenses (**CC0 1.0, CC-BY 4.0, or CC-BY-NC 4.0**) to legally govern reuse and citation credit.

# IPT Workflow · Step 4: Visibility & Registration

Governing dataset access: from private local staging to endorsed public discovery.




Public Resource: Registered with GBIF and discoverable worldwide

### Private (Draft Mode)

Accessible only to authenticated resource managers. Ideal for internal verification, data cleaning, and co-author review.

### Public (Direct Link)

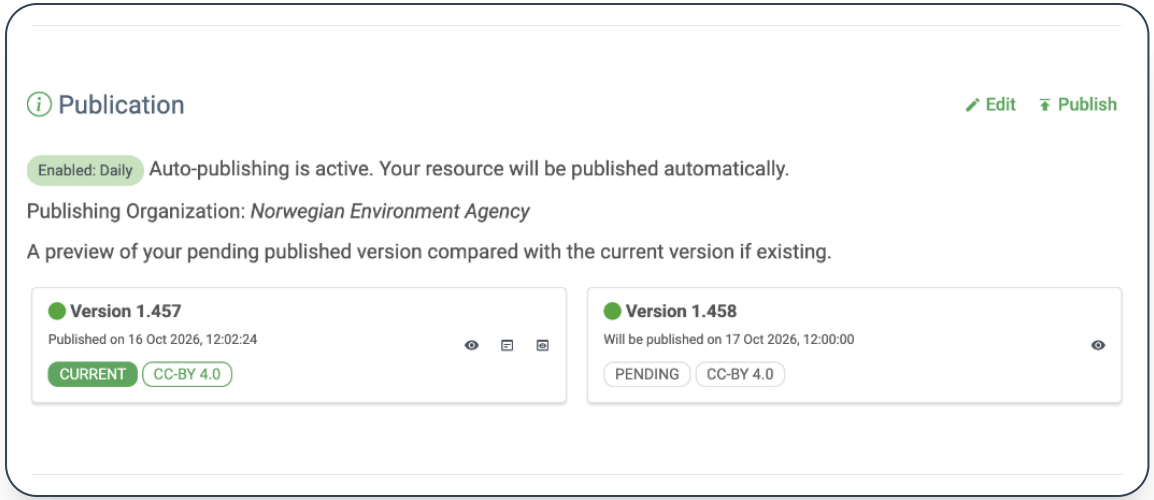
Anyone with the IPT resource URL can download the generated DwC-A archive and read metadata, but not yet indexed on GBIF.org.

### Registered (GBIF Network)

Endorsed by a **Participant Node** and registered in the GBIF Registry. Automatically indexed and citable worldwide.

# IPT Workflow · Step 5: Publication & Versioning

Publishing major/minor versions, minting DOIs, and scheduling automated background sync.




Version Control: Live version 1.457 vs. scheduled daily publication 1.458

### Automated Re-publishing

As shown here (`Enabled: Daily`), databases update records automatically every day without human intervention!

### Semantic Versioning & DOIs

Each major release receives an immutable version tag (`1.457 → 1.458`) and DOI tracking, keeping scientific citations fully reproducible.

### Instant Package Generation

Compiles the ZIP archive containing `meta.xml`, `eml.xml`, and standardized DwC tables for GBIF harvester crawling.

# IPT Workflow · Step 6: GBIF Registration

Attaching the dataset to an endorsed publishing institution and minting global registry keys.




GBIF Registration: Linked UUID key under Norwegian Environment Agency

### 🏛️ Publishing Organization

Links the dataset to an endorsed publisher (e.g. *Norwegian Environment Agency*) approved by a GBIF Participant Node.

### 🔑 GBIF Dataset UUID

Assigns an immutable global UUID (`56fb17ca-adb2...`) in the global GBIF Registry.

### 🤖 Automated Harvester Hook

Tells the central GBIF crawler where to retrieve updates whenever a new version is published in this IPT.

# IPT Workflow · Step 7: Thematic Networks

Connecting datasets to specialized global initiatives (e.g. OBIS, VertNet, national consortia).




GBIF Networks: Grouping datasets into collaborative thematic umbrellas

### Thematic Aggregation

Allows associating the resource with specialized networks like **OBIS** (marine), **VertNet** (vertebrates), or national monitoring programs.

### Cross-Dataset Portals

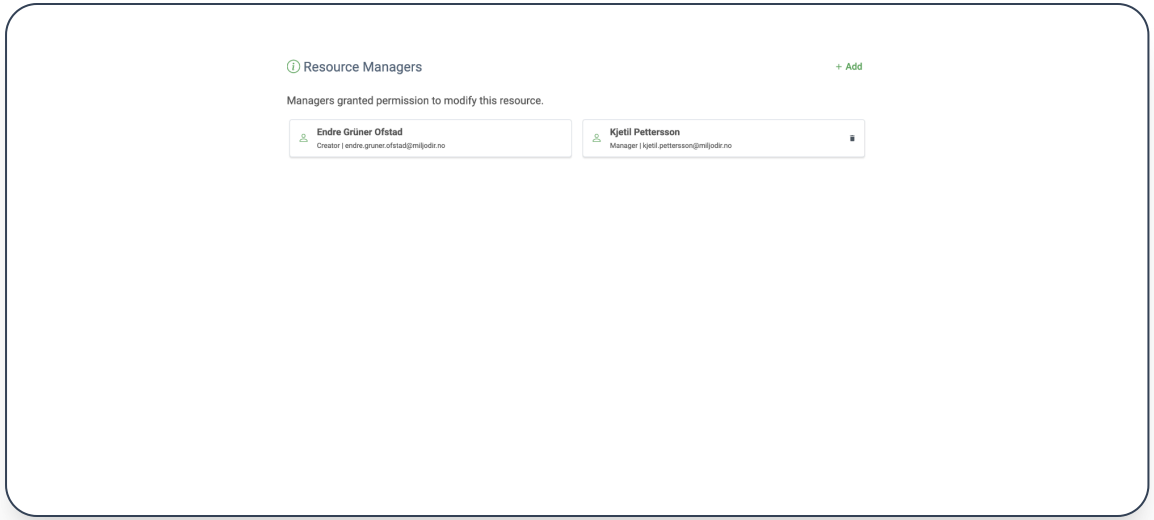
Enables custom filtering and unified analytics across multi-institutional consortia publishing through different IPTs.

### Optional Participation

Resources remain fully published and indexed in GBIF even if no specific secondary network is assigned.

# IPT Workflow · Step 8: Resource Managers

Role-based access control, collaborative dataset curation, and institutional longevity.




Resource Managers: Role-based access control and collaborative curation

### 👥 Collaborative Governance

Allows primary authors and data curators (e.g. *Endre Grüner Ofstad* and *Kjetil Pettersson*) to co-manage the dataset.

### 🔐 Role-Based Access Control

Differentiates between **Resource Creator**, **Manager** (editing mappings/metadata), and institutional **Admins**.

### 🏛️ Institutional Continuity

Prevents orphaned datasets when individual researchers move institutions — multiple curators ensure long-term maintenance.



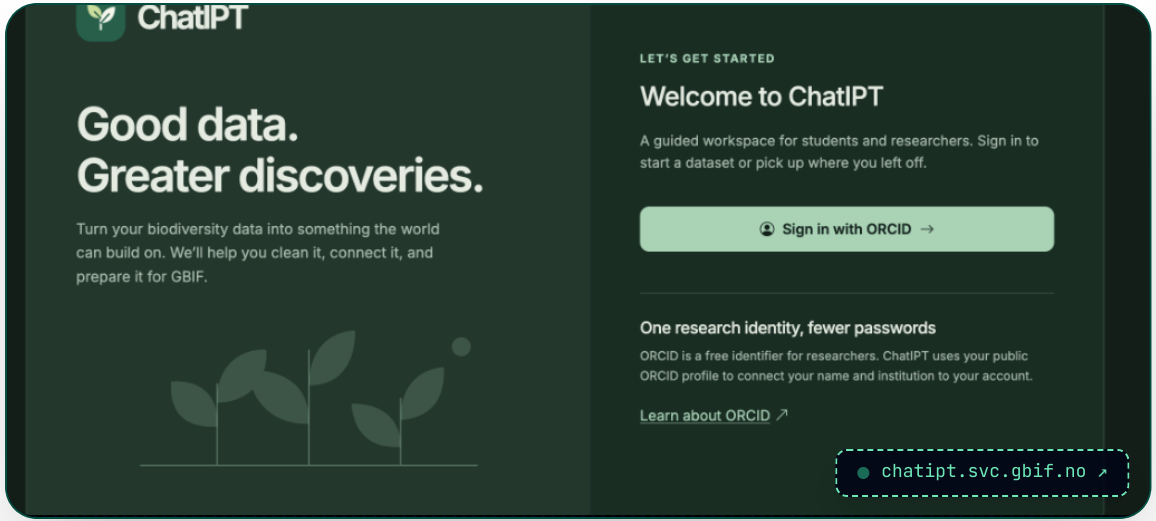
# *"OK, but that still sounds like a lot of work..."*

## ...Can AI help me somehow?

- Yes! From Darwin Core term mapping to drafting EML metadata and validating records.

# Meet ChatIPT: AI-Guided Biodiversity Publishing

A modern, guided AI workspace designed for students and researchers to clean, connect, and prepare datasets for GBIF · chatipt.svc.gbif.no ↗




### 🌱 Good Data, Greater Discoveries

Guided workspace turning raw spreadsheets into standard, connected biodiversity data.

### 🤖 AI-Assisted Cleaning & Mapping

Auto-detects column types, recommends Darwin Core terms, and flags spatial errors.

### 🪪 ORCID-First Authentication

Instant ORCID sign-in tying researcher identity, provenance, and citations directly to datasets.

**Launch ChatIPT** chatipt.svc.gbif.no ↗

# Hands-On Lab: Let’s Try It!

Put biodiversity data publishing into practice — choose your track:

CLASSIC WORKFLOW

## GBIF IPT (Sandbox)

The institutional workhorse hosted by national participant nodes for comprehensive dataset administration.

- ✓ Connect raw CSV, TSV, or live SQL databases
- ✓ Map columns directly to Darwin Core Core & Extensions
- ✓ Fill rich EML metadata & manage institutional roles

Recommended for data managers

Open IPT ↗

AI-GUIDED WORKFLOW

## ChatIPT Workspace

Intelligent assistant turning raw researcher spreadsheets into validated, connected GBIF datasets.

- ✓ 1-click ORCID authentication (no passwords)
- ✓ AI automated column detection & DwC term suggestions
- ✓ Interactive coordinate cleaning & instant package generation

Recommended for students & researchers

Launch ChatIPT ↗

🎯 **Hands-on Goal:** Take your own sample spreadsheet (or our provided demo CSV), standardize it, and create your first valid Darwin Core dataset!

Hands-on Lab · Lecture 01